LESSON 01 / 10

Coordinates locate; displacements direct

A point $P=(x,y,z)$ is located by three signed coordinates. The coordinate planes $x=0$, $y=0$ and $z=0$ divide space into eight octants. Reflection in the xy-plane changes only the sign of z; reflection in the origin reverses all three coordinates.

For $A=(x_1,y_1,z_1)$ and $B=(x_2,y_2,z_2)$, the displacement and distance are

$$\vec{AB}=(x_2-x_1,y_2-y_1,z_2-z_1),$$

$$AB=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}.$$

The displacement supplies direction ratios for the line through A,B, provided $A\neq B$.

WORKED EXAMPLE

Distance and displacement are different answers

For $A=(1,2,3)$ and $B=(4,6,9)$,

$$\vec{AB}=(3,4,6),\qquad AB=\sqrt{9+16+36}=\sqrt{61}.$$

The direction ratios are $3:4:6$; the number $\sqrt{61}$ is a length, not a direction.

Direction cosines are components of a unit direction

A nonzero direction vector $\mathbf{d}=(a,b,c)$ has direction ratios $a:b:c$. Normalize it to obtain the direction cosines

$$(l,m,n)=\hat{\mathbf{d}}=\frac{(a,b,c)}{\sqrt{a^2+b^2+c^2}},\qquad l^2+m^2+n^2=1.$$

Here l,m,n are the cosines of the angles that the directed line makes with the positive x,y,z axes. Reversing the direction reverses all three cosines.

WORKED EXAMPLE

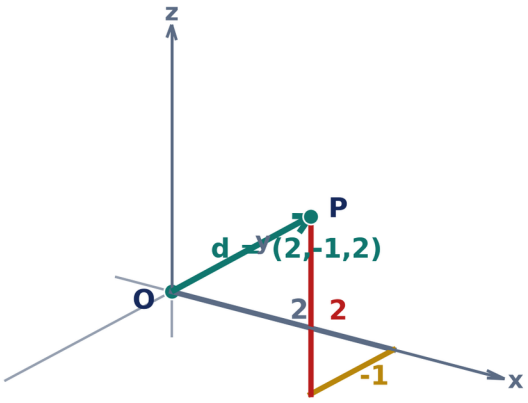
Normalize before calling ratios cosines

For $\mathbf{d}=(2,-1,2)$, $|\mathbf{d}|=3$, so

$$(l,m,n)=\left(\tfrac{2}{3},-\tfrac{1}{3},\tfrac{2}{3}\right),\qquad \tfrac{4}{9}+\tfrac{1}{9}+\tfrac{4}{9}=1.$$

The original triple $(2,-1,2)$ has squared sum 9, so it cannot itself be a direction-cosine triple.

Direction cosines come from the normalized displacement



READ THE FIGURE
The component path has signed steps 2,-1,2. Its resultant has length 3 and unit direction (2/3,-1/3,2/3).

Projection uses a unit direction

The signed projection of $\vec{AB}$ on a line with nonzero direction $\mathbf{d}$ is

$$\vec{AB}\cdot\hat{\mathbf{d}}=\frac{\vec{AB}\cdot\mathbf{d}}{|\mathbf{d}|}.$$

It is positive when the displacement has a component along the chosen direction and negative when the component opposes it.

WORKED EXAMPLE
A segment projected onto another direction
Using $\vec{AB}=(3,4,6)$ and $\mathbf{d}=(2,-1,2)$,

$$\text{projection}=\frac{6-4+12}{3}=\frac{14}{3}.$$

For direction cosines $(1/2,1/\sqrt{2},1/2)$, the angle with the positive z-axis is $\gamma=60^\circ$ because $n=1/2$.

COMMON TRAP
Zero has no direction
Never normalize $(0,0,0)$. Two coincident points do not determine a line, and a zero direction ratio triple does not define direction cosines or an angle.

LESSON 02 / 10

One point and one nonzero direction determine a line

Through A with position vector $\mathbf{a}$ and direction $\mathbf{b} \neq \mathbf{0}$, every point of the line is

$$\mathbf{r} = \mathbf{a} + t\mathbf{b}, \qquad t \in \mathbb{R}.$$

If $\mathbf{a} = (x_1, y_1, z_1)$ and $\mathbf{b} = (a, b, c)$, the parametric equations are

$$x = x_1 + at, \qquad y = y_1 + bt, \qquad z = z_1 + ct.$$

When a, b, c are all nonzero, eliminate t to obtain symmetric form:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}.$$

For a second point B, use $\mathbf{b} = \overrightarrow{AB}$; this gives the two-point form.

WORKED EXAMPLE

Two points become one vector equation

For $A = (1, 2, 3)$ and $B = (4, 6, 9)$, $\overrightarrow{AB} = (3, 4, 6)$. Hence

$$\mathbf{r} = (1, 2, 3) + t(3, 4, 6),$$

or

$$\frac{x - 1}{3} = \frac{y - 2}{4} = \frac{z - 3}{6}.$$

At $t = 1$ the point is B; at $t = 1/2$ it is the midpoint $(5/2, 4, 6)$.

COMMON TRAP

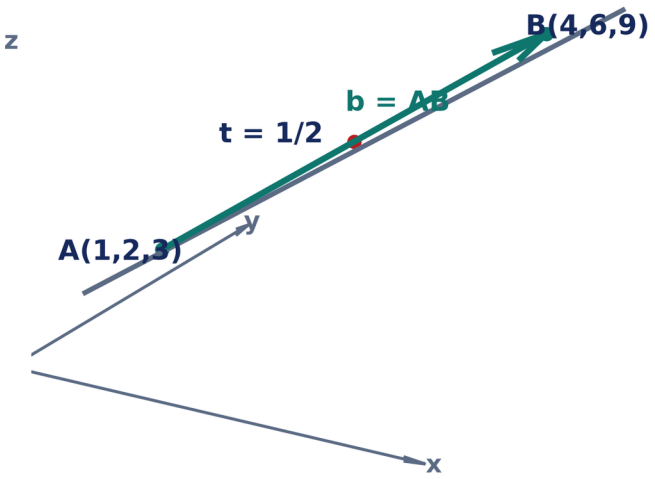
A zero denominator needs a coordinate equation

Through $(1, 2, 3)$ parallel to the z-axis, write

$$x = 1, \qquad y = 2, \qquad z = 3 + t.$$

Do not place zero under $x - 1$ or $y - 2$ in the symmetric formula. The vector and parametric forms remain valid without special cases.

A line is one point plus all scalar multiples of a direction



READ THE FIGURE
The segment AB supplies the direction, but the line continues beyond both endpoints. Parameter values locate every point on it.

Two independent planes also describe a line

The intersection of nonparallel planes

$$P_1 = n_{11}x + n_{12}y + n_{13}z - d_1 = 0, \qquad P_2 = n_{21}x + n_{22}y + n_{23}z - d_2 = 0$$

is a line. Its direction is

$$\mathbf{b} = \mathbf{n}_1 \times \mathbf{n}_2,$$

because a vector along both planes must be perpendicular to both normals. Find one point by assigning a convenient coordinate and solving the remaining two equations.

WORKED EXAMPLE

From general form to symmetric form

For

$$x+y+z=1, \qquad 2x-3y+4z=5,$$

the normals give

$$(1,1,1)\times(2,-3,4)=(7,-2,-5).$$

Set $z=0$. Then $x+y=1$ and $2x-3y=5$, so the point is

$$P=\left(\tfrac{8}{5},-\tfrac{3}{5},0\right).$$

Thus

$$\frac{x-8/5}{7}=\frac{y+3/5}{-2}=\frac{z}{-5}.$$

Substitution checks both plane equations, and $(7,-2,-5)$ dots both normals to zero.

KEY RESULT

Convert only after checking independence

If $\mathbf{n}_1\times\mathbf{n}_2=\mathbf{0}$, the plane normals are parallel: the planes may be parallel distinct or coincident, but they do not determine a unique intersection line. Compute the cross product before hunting for a point.

LESSON 03 / 10

Directions determine the angle, not the points

For lines with nonzero directions $\mathbf{b}_1, \mathbf{b}_2$, the acute angle θ between the lines satisfies

$$\cos\theta = \frac{|\mathbf{b}_1 \cdot \mathbf{b}_2|}{|\mathbf{b}_1||\mathbf{b}_2|}, \qquad 0 \le \theta \le \frac{\pi}{2}.$$

The absolute value removes the arbitrary choice of arrow orientation. Perpendicular lines have dot product 0; parallel lines have cross product $\mathbf{0}$.

WORKED EXAMPLE

An exact angle between two directions

For $\mathbf{b}_1 = (1, 2, 2)$ and $\mathbf{b}_2 = (2, 1, -1)$,

$$\mathbf{b}_1 \cdot \mathbf{b}_2 = 2, \qquad |\mathbf{b}_1| = 3, \quad |\mathbf{b}_2| = \sqrt{6}.$$

Therefore

$$\theta = \cos^{-1}\left(\frac{2}{3\sqrt{6}}\right) \approx 74.21^\circ.$$

For directions $(2, 3, 4)$ and $(4, 3, 2)$, the dot product is $25 \neq 0$, so those directions are not perpendicular.

Coplanarity is a scalar-triple-product test

Let $L_1 : \mathbf{r} = \mathbf{a}_1 + t\mathbf{b}_1$ and $L_2 : \mathbf{r} = \mathbf{a}_2 + s\mathbf{b}_2$ be nonparallel. They are coplanar exactly when

$$(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{b}_1 \times \mathbf{b}_2) = 0.$$

In three dimensions, nonparallel coplanar lines intersect. If the triple product is nonzero, the lines are **skew**: they neither intersect nor lie in one plane. Parallel lines are always coplanar and need a separate coincident-versus-distinct test.

WORKED EXAMPLE

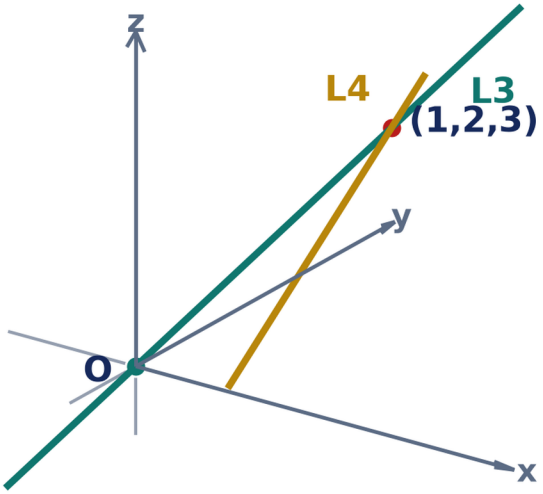
A zero or nonzero test with a geometric conclusion

For $L_1 : \mathbf{r} = t(1, 0, 0)$ and $L_2 : \mathbf{r} = (0, 1, 0) + s(0, 0, 1)$,

$$(0, 1, 0) \cdot ((1, 0, 0) \times (0, 0, 1)) = -1 \neq 0.$$

Thus the lines are skew. For $L_3 : \mathbf{r} = t(1, 2, 3)$ and $L_4 : \mathbf{r} = (1, 0, 1) + s(0, 1, 1)$, the same test is zero. Solving gives $t = 1$, $s = 2$ and the intersection $(1, 2, 3)$.

Nonparallel coplanar lines meet at one point



READ THE FIGURE
L3=t(1,2,3) and L4=(1,0,1)+s(0,1,1) meet when t=1 and s=2. Their directions are not parallel.

Intersection means solving for parameters, not guessing

Equate the three coordinates of $\mathbf{a}_1 + t\mathbf{b}_1$ and $\mathbf{a}_2 + s\mathbf{b}_2$. Two equations usually determine t, s; the third is the essential consistency check. If the system is inconsistent, the lines do not intersect.

COMMON TRAP
Parallel directions do not distinguish coincident from distinct
For

$$L_1 : \mathbf{r} = t(1, 2, 3), \qquad L_2 : \mathbf{r} = (1, 2, 3) + s(1, 2, 3),$$

the directions are identical, but $(1, 2, 3)$ also lies on L_1 at $t = 1$. Therefore the lines **coincide**. Replacing the second base point by $(1, 0, 0)$ would make them distinct and parallel. Always test the base point.

KEY RESULT
Classify in a fixed order
First test directions: parallel or not? If parallel, test coincidence. If nonparallel, compute the coplanarity triple product. Zero means an intersection; nonzero means skew lines and leads to the shortest-distance formula.

LESSON 04 / 10

The foot is the perpendicular projection

For $L : \mathbf{r} = \mathbf{a} + t\mathbf{b}$ and a point P with position vector $\mathbf{p}$, let $F = \mathbf{a} + t_0\mathbf{b}$ be the foot of the perpendicular. The condition $(P - F) \cdot \mathbf{b} = 0$ gives

$$t_0 = \frac{(\mathbf{p} - \mathbf{a}) \cdot \mathbf{b}}{|\mathbf{b}|^2}, \qquad F = \mathbf{a} + t_0\mathbf{b}.$$

The image of P in the line is $I = 2F - P$, because F is the midpoint of PI. The point-line distance is simply $|P - F|$.

SHORT DERIVATION

The same parameter minimizes distance

Expanding $|P - (\mathbf{a} + t\mathbf{b})|^2$ gives a quadratic in t. Its minimum occurs when the derivative is zero, which is exactly $(P - F) \cdot \mathbf{b} = 0$. Thus nearest point, perpendicular foot and orthogonal projection are three descriptions of the same point.

WORKED EXAMPLE

Foot, image and distance in one calculation

Take $P = (1, 2, 3)$ and $L : \mathbf{r} = t(2, -1, 2)$. Since

$$t_0 = \frac{(1, 2, 3) \cdot (2, -1, 2)}{9} = \frac{2}{3},$$

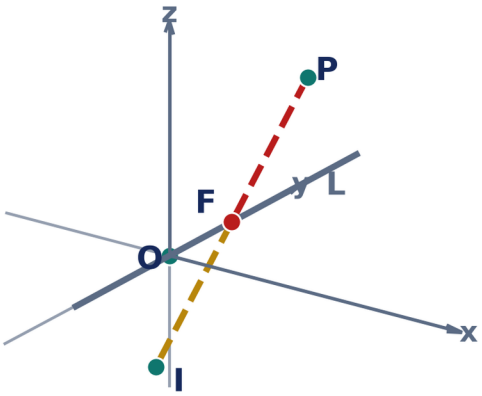
the foot and image are

$$F = \left(\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right), \qquad I = 2F - P = \left(\frac{5}{3}, -\frac{10}{3}, -\frac{1}{3}\right).$$

Also $P - F = (-1/3, 8/3, 5/3)$, whose dot product with $(2, -1, 2)$ is zero. Hence

$$PF = \sqrt{\frac{1 + 64 + 25}{9}} = \sqrt{10}.$$

The foot bisects the point and its reflected image



READ THE FIGURE

L=t(2,-1,2). PF is perpendicular to L and F is the midpoint of PI. The rounded plotted F is (4/3,-2/3,4/3).

A cross product computes the distance without finding F

The perpendicular distance from P to L is also

$$d = \frac{|(\mathbf{p}-\mathbf{a})\times\mathbf{b}|}{|\mathbf{b}|}.$$

The numerator is the parallelogram area with base $|\mathbf{b}|$; dividing by the base leaves the height. This is useful when only the distance is requested.

WORKED EXAMPLE

Check the same distance by area

For the example above, $\mathbf{p}-\mathbf{a}=(1,2,3)$ and $\mathbf{b}=(2,-1,2)$:

$$(1,2,3)\times(2,-1,2)=(7,4,-5).$$

Therefore

$$d=\frac{\sqrt{49+16+25}}{3}=\frac{\sqrt{90}}{3}=\sqrt{10}.$$

This agrees with $|P-F|$ and avoids rounding the foot.

KEY RESULT

Compute only what the question needs

Use t_0 when the foot or image is required. Use the cross-product height when only the distance is required. In either method, finish with an independent perpendicularity or length check.

LESSON 05 / 10

Shortest distance is volume divided by base area

For nonparallel lines

$$L_1 : \mathbf{r} = \mathbf{a}_1 + t\mathbf{b}_1, \qquad L_2 : \mathbf{r} = \mathbf{a}_2 + s\mathbf{b}_2,$$

their common-perpendicular direction is $\mathbf{b}_1 \times \mathbf{b}_2$. The shortest distance is

$$d = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{b}_1 \times \mathbf{b}_2)|}{|\mathbf{b}_1 \times \mathbf{b}_2|}.$$

The numerator is a parallelepiped volume and the denominator is the area of the base spanned by the two directions. Their quotient is the perpendicular height.

If the numerator is zero, the lines are coplanar and intersect because they are nonparallel. A positive value means genuinely skew lines.

WORKED EXAMPLE

A complete skew-line distance

For

$$L_1 : (1, 2, 3) + t(2, -1, 2), \qquad L_2 : (4, 5, 6) + s(3, 1, -1),$$

compute

$$\mathbf{b}_1 \times \mathbf{b}_2 = (-1, 8, 5), \qquad \mathbf{a}_2 - \mathbf{a}_1 = (3, 3, 3).$$

Thus the signed numerator is $-3 + 24 + 15 = 36$ and

$$d = \frac{36}{\sqrt{90}} = \frac{12}{\sqrt{10}} \approx 3.7947.$$

The nonzero cross product and numerator independently confirm that the lines are skew.

The common perpendicular can be located exactly

Let $F_1 = \mathbf{a}_1 + t\mathbf{b}_1$ and $F_2 = \mathbf{a}_2 + s\mathbf{b}_2$ be its feet. Solve

$$(F_2 - F_1) \cdot \mathbf{b}_1 = 0, \qquad (F_2 - F_1) \cdot \mathbf{b}_2 = 0.$$

For the same lines, these equations become

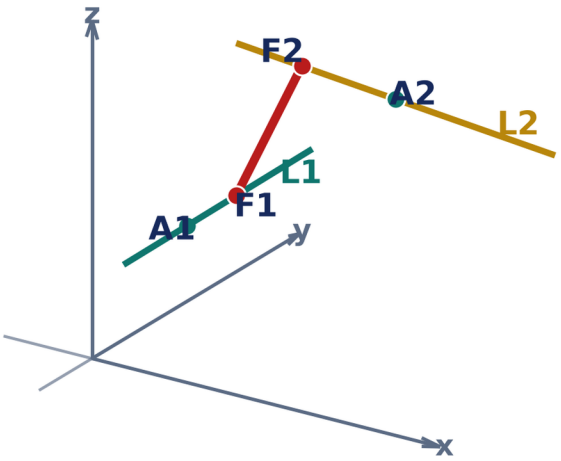
$$9 + 3s - 9t = 0, \qquad 9 + 11s - 3t = 0,$$

so $t = 4/5$ and $s = -3/5$. Therefore

$$F_1 = \left(\tfrac{13}{5}, \tfrac{6}{5}, \tfrac{23}{5}\right), \qquad F_2 = \left(\tfrac{11}{5}, \tfrac{22}{5}, \tfrac{33}{5}\right).$$

Indeed $F_2 - F_1 = (2/5)(-1, 8, 5)$, whose length is $(2/5)\sqrt{90} = 12/\sqrt{10}$.

The common perpendicular is parallel to b1 × b2



READ THE FIGURE
F1=(13/5,6/5,23/5), F2=(11/5,22/5,33/5), and F2-F1=(2/5)(-1,8,5). The red connector is perpendicular to both lines.

Do not divide by a zero cross product

The skew-line formula requires nonparallel directions. For parallel lines, choose a point A_2 on the second line and compute its distance to the first:

$$d = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \times \mathbf{b}_1|}{|\mathbf{b}_1|}.$$

For example, the x-axis and $(0, 2, 0) + s(1, 0, 0)$ are distinct parallel lines with distance 2. Coincident lines have distance 0.

KEY RESULT
The numerator and denominator answer different questions
A zero denominator means parallel directions. A zero numerator with nonzero denominator means coplanar intersecting lines. Both nonzero means skew lines with positive shortest distance. Classify before substituting.

LESSON 06 / 10

One point and one nonzero normal determine a plane

The plane through A with normal $\mathbf{n} \neq \mathbf{0}$ is

$$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0.$$

If $\mathbf{n} = (n_1, n_2, n_3)$, this becomes

$$n_1 x + n_2 y + n_3 z = d, \qquad d = \mathbf{a} \cdot \mathbf{n}.$$

Multiplying the entire equation by a nonzero constant does not change the plane. The normal, however, must remain nonzero.

WORKED EXAMPLE

Compute the constant from the given point

Through $A = (1, 2, 3)$ with normal $(2, -3, 4)$,

$$d = 2 - 6 + 12 = 8,$$

so the plane is

$$2x - 3y + 4z = 8.$$

Substitution of A gives 8, while the normal dots every displacement inside the plane to zero.

Three noncollinear points supply a normal

For A, B, C noncollinear, take two edge vectors and cross them:

$$\mathbf{n} = (B - A) \times (C - A).$$

Then use any one of the three points to compute d. A zero cross product would mean collinear points, which do not determine a unique plane.

WORKED EXAMPLE

Three intercepts become one plane equation

For $A = (1, 0, 0)$, $B = (0, 2, 0)$ and $C = (0, 0, 3)$,

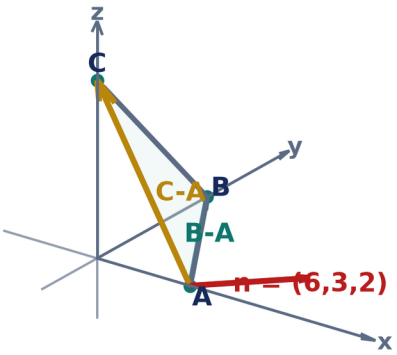
$$(-1, 2, 0) \times (-1, 0, 3) = (6, 3, 2), \qquad d = (6, 3, 2) \cdot (1, 0, 0) = 6.$$

Thus

$$6x + 3y + 2z = 6 \iff x + \tfrac{y}{2} + \tfrac{z}{3} = 1.$$

The second form is the intercept form $x/a + y/b + z/c = 1$, here with $a = 1, b = 2, c = 3$.

Two edges fix the plane; their cross product fixes its normal



READ THE FIGURE
The normal drawn from A is a scaled (6,3,2). All three vertices satisfy 6x+3y+2z=6, equivalently x+y/2+z/3=1.

Intercepts and normals are reciprocal descriptions

For nonzero intercepts a, b, c, the plane

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

has normal $(1/a, 1/b, 1/c)$, or any nonzero scalar multiple. If an intercept is infinite or zero, do not force this form; return to the normal equation.

KEY RESULT
Membership and perpendicularity are the two fastest checks
After finding a plane, substitute every required point. After finding a line in a plane, check that its direction dots the plane's normal to zero. These checks catch a wrong constant and a wrong normal independently.

LESSON 07 / 10

Plane–plane angle uses the normals

For planes with nonzero normals $\mathbf{n}_1, \mathbf{n}_2$, the acute angle θ between the planes satisfies

$$\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|}.$$

Parallel normals give parallel or coincident planes. Perpendicular normals give perpendicular planes. An oriented dihedral angle may require a sign convention, but the standard acute angle uses the absolute value.

WORKED EXAMPLE

Angle between two planes

For $x + y + z = 0$ and $x - y + z = 0$, the normals are $(1, 1, 1)$ and $(1, -1, 1)$. Their dot product is 1 and both lengths are $\sqrt{3}$, so

$$\cos\theta = \tfrac{1}{3}, \qquad \theta = \cos^{-1}\left(\tfrac{1}{3}\right) \approx 70.53^\circ.$$

Line–plane angle is complementary to line–normal angle

For a line with direction $\mathbf{b}$ and a plane with normal $\mathbf{n}$, the acute angle θ between the line and plane satisfies

$$\sin\theta = \frac{|\mathbf{b} \cdot \mathbf{n}|}{|\mathbf{b}||\mathbf{n}|}.$$

If ϕ is the acute angle between $\mathbf{b}$ and $\mathbf{n}$, then $\theta = 90^\circ - \phi$. A line parallel to the plane has $\mathbf{b} \cdot \mathbf{n} = 0$ and $\theta = 0$; a line perpendicular to the plane has $\mathbf{b} \parallel \mathbf{n}$ and $\theta = 90^\circ$.

WORKED EXAMPLE

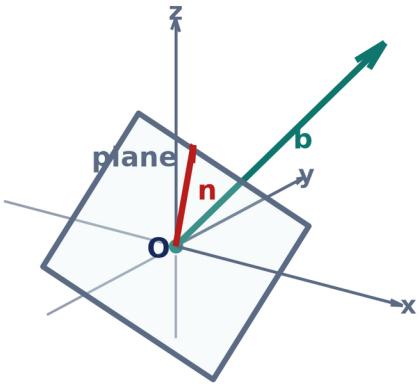
Use sine, then verify by complement

For $\mathbf{b} = (2, 3, 4)$ and the plane $2x - 3y + 4z = 5$, the normal is $\mathbf{n} = (2, -3, 4)$. Both lengths are $\sqrt{29}$, while $\mathbf{b} \cdot \mathbf{n} = 11$. Hence

$$\sin\theta = \tfrac{11}{29}, \qquad \theta \approx 22.29^\circ.$$

The line–normal angle is $\phi = \cos^{-1}(11/29) \approx 67.71^\circ$, and indeed $\theta + \phi = 90^\circ$.

Line-plane angle is the complement of line-normal angle



READ THE FIGURE

b=(2,3,4), n=(2,-3,4), and the plane is 2x-3y+4z=0. The displayed angle formula gives sin(theta)=11/29.

The intersection direction is perpendicular to both normals

For nonparallel planes, their line of intersection has direction $\mathbf{n}_1 \times \mathbf{n}_2$. This also explains why a line lying in both planes must have zero dot product with each normal.

COMMON TRAP

Do not use the line–line formula for a line and a plane

The line–plane formula contains $\sin\theta$ because θ is measured from the plane, not from its normal. Using $\cos\theta$ there gives the complementary angle.

KEY RESULT

Second read: cosine and tangent of the same angle

For the worked line–plane example,

$$\cos\theta = \sqrt{1-\left(\frac{11}{29}\right)^2} = \frac{12\sqrt{5}}{29}, \qquad \tan\theta = \frac{11}{12\sqrt{5}}.$$

These follow from the same right triangle and provide an independent exact check.

Distance is the absolute normalized residual

For P with position vector $\mathbf{p}$ and plane $\mathbf{n}\cdot\mathbf{r}=d$,

$$\operatorname{dist}(P,\text{plane})=\frac{|\mathbf{p}\cdot\mathbf{n}-d|}{|\mathbf{n}|}.$$

The numerator measures how far P is from satisfying the equation; division by $|\mathbf{n}|$ converts that algebraic residual to a length.

For parallel planes $\mathbf{n}\cdot\mathbf{r}=d_1$ and $\mathbf{n}\cdot\mathbf{r}=d_2$ with the **same** normal coefficients,

$$d_{\text{planes}}=\frac{|d_1-d_2|}{|\mathbf{n}|}.$$

If one equation is scaled differently, normalize or rewrite it first.

WORKED EXAMPLE

Point distance and parallel-plane distance

For $P=(1,1,1)$ and $2x-3y+4z=5$,

$$\frac{|2-3+4-5|}{\sqrt{29}}=\frac{2}{\sqrt{29}}.$$

The distance between $2x-3y+4z=5$ and $2x-3y+4z=10$ is

$$\frac{|10-5|}{\sqrt{29}}=\frac{5}{\sqrt{29}}.$$

Foot and image move along the normal

Write the foot as $F=P+t\mathbf{n}$ and impose $\mathbf{n}\cdot F=d$. This yields

$$F=P-\frac{\mathbf{p}\cdot\mathbf{n}-d}{|\mathbf{n}|^2}\,\mathbf{n},\qquad I=2F-P.$$

The image I is the reflection of P in the plane; F is the midpoint of PI, and PI is perpendicular to the plane.

WORKED EXAMPLE

Reflection in x+y+z=1

For $P=(1,2,3)$ and $\mathbf{n}=(1,1,1)$,

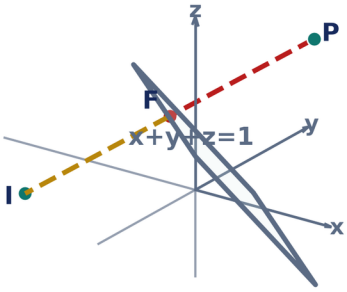
$$\frac{\mathbf{p}\cdot\mathbf{n}-d}{|\mathbf{n}|^2}=\frac{6-1}{3}=\frac{5}{3}.$$

Therefore

$$F=\left(-\tfrac{2}{3},\tfrac{1}{3},\tfrac{4}{3}\right),\qquad I=\left(-\tfrac{7}{3},-\tfrac{4}{3},-\tfrac{1}{3}\right).$$

The foot satisfies $x+y+z=1$, and $P+I=2F$.

Reflection in a plane is symmetric about the perpendicular foot



READ THE FIGURE

The plotted points use the exact values $P=(1,2,3)$, $F=(-2/3,1/3,4/3)$, $I=(-7/3,-4/3,-1/3)$, rounded only for rendering.

WORKED EXAMPLE

A second foot with a distance check

For the same P and $2x-3y+4z=5$,

$$\frac{\mathbf{p}\cdot\mathbf{n}-d}{|\mathbf{n}|^2}=\frac{3}{29},$$

so

$$F=\left(\frac{23}{29},\frac{67}{29},\frac{75}{29}\right).$$

Substitution gives $2F_x-3F_y+4F_z=5$, and $PF=3/\sqrt{29}$.

KEY RESULT

The residual controls all three answers

The same signed quantity $\mathbf{p}\cdot\mathbf{n}-d$ gives the distance after normalization, the foot after multiplication by $\mathbf{n}/|\mathbf{n}|^2$, and the image after doubling the displacement from P to F.

LESSON 09 / 10

A pencil keeps the common line automatically

Suppose $P_1 = 0$ and $P_2 = 0$ are independent plane equations. Every plane of the form

$$P_1 + \lambda P_2 = 0$$

contains their line of intersection, because both residuals vanish there. This one-parameter form covers every plane through that line **except** $P_2 = 0$ **itself**. To include every member at once, use the homogeneous pencil

$$\alpha P_1 + \beta P_2 = 0, \qquad (\alpha, \beta) \neq (0, 0).$$

Impose one additional point or normal condition to find the required member.

WORKED EXAMPLE

A plane through an intersection line and the origin

For

$$P_1 = x + y + z - 1, \qquad P_2 = 2x - 3y + 4z - 5,$$

use $P_1 + \lambda P_2 = 0$. At the origin, $-1 - 5\lambda = 0$, so $\lambda = -1/5$. Multiplying by 5 gives

$$3x + 8y + z = 0.$$

This contains the origin and every point satisfying both original plane equations.

WORKED EXAMPLE

A second point determines a second pencil member

Through the intersection of $x + 2y + 3z = 4$ and $2x + y - z = 5$, and through $(1, 1, 1)$, substitution into the pencil gives $\lambda = 2/3$. Clearing denominators,

$$7x + 8y + 7z = 22.$$

The required point and the common line satisfy this equation.

Bisector planes are equal-distance loci

For planes $P_1 = 0$ and $P_2 = 0$ with normals $\mathbf{n}_1, \mathbf{n}_2$, equal distances to the two planes require

$$\frac{|P_1|}{|\mathbf{n}_1|} = \frac{|P_2|}{|\mathbf{n}_2|}.$$

Removing absolute values gives the two bisector planes:

$$\frac{P_1}{|\mathbf{n}_1|} = \pm \frac{P_2}{|\mathbf{n}_2|}.$$

The two signs select the two angle bisectors; normalization is essential when the normal lengths differ.

WORKED EXAMPLE

Two perpendicular bisector planes

For $x+y+z=0$ and $x-y+z=0$, both normal lengths are $\sqrt{3}$. Hence

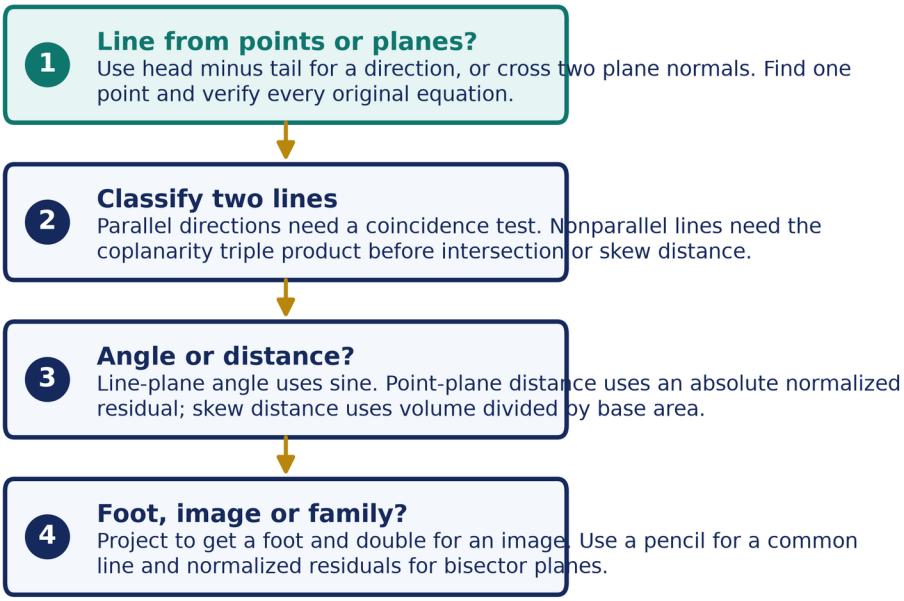
$$x+y+z=\pm(x-y+z).$$

The two planes are

$$y=0 \qquad \text{and} \qquad x+z=0.$$

Their normals $(0,1,0)$ and $(1,0,1)$ have dot product zero, so the bisector planes are perpendicular.

Choose the smallest sufficient 3D computation



READ THE FIGURE

Name the geometric object first. Then select a formula whose hypotheses match it and finish with direct substitution or perpendicularity.

COMMON TRAP

Do not normalize only part of a formula

Direction ratios must be normalized for direction cosines, plane residuals for equal-distance loci, and parallel-plane equations before subtracting constants. Keep exact coefficients until the correct normalization is clear.

KEY RESULT

Final 3D self-check

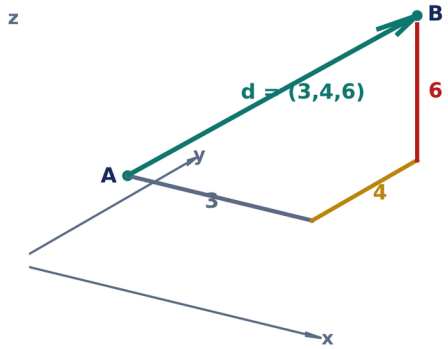
Does the point satisfy every required equation? Is every direction or normal nonzero? Does the line direction dot each relevant normal to zero? Are the common-perpendicular connector and the two line directions mutually perpendicular? Does the requested angle use cosine or sine appropriately? These checks are faster than repeating the same algebra.

LESSON 10 / 10

A displacement becomes a direction after normalization

For $A=(1,2,3)$ and $B=(4,6,9)$, the edge vector is $\mathbf{d}=(3,4,6)$ and $|\mathbf{d}|=\sqrt{61}$. Its direction cosines are therefore $(3,4,6)/\sqrt{61}$.

The component path and the direct displacement have the same endpoints



READ THE FIGURE

The three component steps have squared lengths 9,16,36. The direct displacement has squared length 61 and unit direction (3,4,6)/sqrt(61).

Visual check: $(3^2+4^2+6^2)/61=1$, so the normalized components are genuine direction cosines.

KEY RESULT

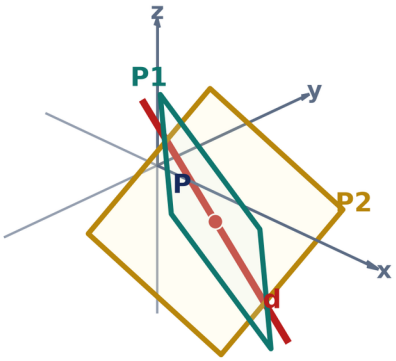
Second read: projection is another normalized dot product

Using the same displacement and the direction $\mathbf{b}=(2,-1,2)$, whose length is 3, the signed projection is $(6-4+12)/3=14/3$. The raw dot product is 14; division by the direction length turns it into a length.

Two planes reveal their intersection direction

For $x+y+z=1$ and $2x-3y+4z=5$, the normals are $\mathbf{n}_1=(1,1,1)$ and $\mathbf{n}_2=(2,-3,4)$. Their cross product is $\mathbf{d}=(7,-2,-5)$, and $P=(8/5,-3/5,0)$ lies on both planes.

The intersection line is perpendicular to both plane normals



READ THE FIGURE

The line is P+t⟨7,-2,-5⟩. Every direction along it has zero dot product with both normals, so it stays in both planes.

Visual check: $(7,-2,-5)\cdot(1,1,1)=0$ and $(7,-2,-5)\cdot(2,-3,4)=14+6-20=0$.

KEY RESULT

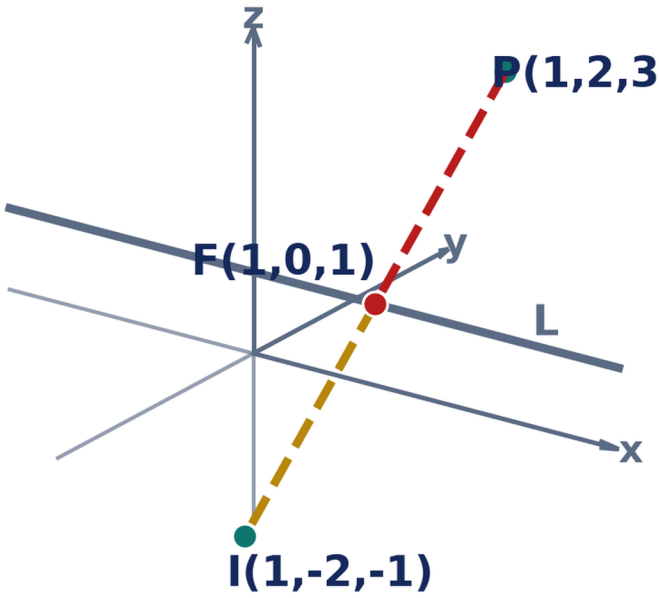
Second read: verify the whole parametrization

At $P+t\mathbf{d}$, both plane residuals equal their values at P plus t times a zero directional dot product. Thus one verified point and two verified perpendicularities prove that the entire parametrized line is the intersection.

A perpendicular foot creates a reflection

Let $L : (0,0,1) + t(1,0,0)$ and $P = (1,2,3)$. Projection gives $F = (1,0,1)$ and reflection gives $I = 2F - P = (1,\ -2,\ -1)$.

F is the midpoint of P and its image I



READ THE FIGURE

PF=(0,-2,-2) is perpendicular to the x-direction. The same displacement continued past F reaches I, so PF=FI=2 sqrt(2).

Visual check: $P + I = (2,0,2) = 2F$, and $(P - F) \cdot (1,0,0) = 0$.

KEY RESULT

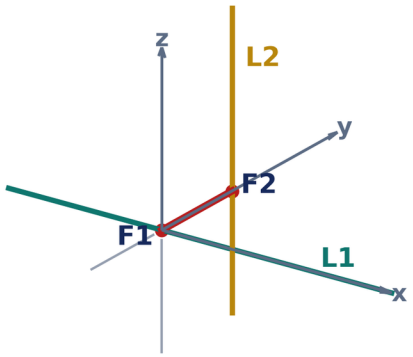
Second read: cross-product height finds the distance without F

With $A = (0,0,1)$ and $\mathbf{b} = (1,0,0)$, $(P - A) \times \mathbf{b} = (0,2,\ -2)$. Dividing its length $2\sqrt{2}$ by $|\mathbf{b}| = 1$ gives the same distance $2\sqrt{2}$.

Skew distance is the gap along a common normal

For $L_1 : \mathbf{r} = t(1, 0, 0)$ and $L_2 : \mathbf{r} = (0, 1, 0) + s(0, 0, 1)$, the directions cross to $(0, -1, 0)$. The connector from $F_1 = (0, 0, 0)$ to $F_2 = (0, 1, 0)$ is perpendicular to both lines.

The shortest connector is perpendicular to both skew lines



READ THE FIGURE

The red connector has length 1. It is antiparallel to b1 × b2=(0,-1,0), and its dot product with each line direction is zero.

Visual check: the signed triple product is $(0, 1, 0) \cdot (0, -1, 0) = -1$, and $d = |-1|/|(0, -1, 0)| = 1$.

KEY RESULT

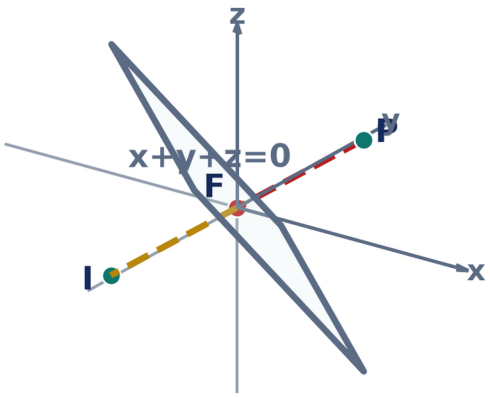
Second read: volume divided by base area

The connector and the two unit directions form a unit parallelepiped of volume 1. The base spanned by the line directions has area 1, so its height is 1. A zero volume would instead put the two lines in one plane.

A plane reflection follows its normal

For the plane $x+y+z=0$ and $P=(1,1,1)$, the foot is $F=(0,0,0)$ and the image is $I=(-1,-1,-1)$. The displacement $P-F=(1,1,1)$ is the plane's normal direction.

P and I are symmetric about the plane through F



READ THE FIGURE

The residual is 3 and the normal squared length is 3, so F=P-(1,1,1)=0 and I=P-2(1,1,1)=(-1,-1,-1).

Visual check: $P+I=2F$ and the point-plane distance is $|3|/\sqrt{3}=\sqrt{3}=|P-F|$.

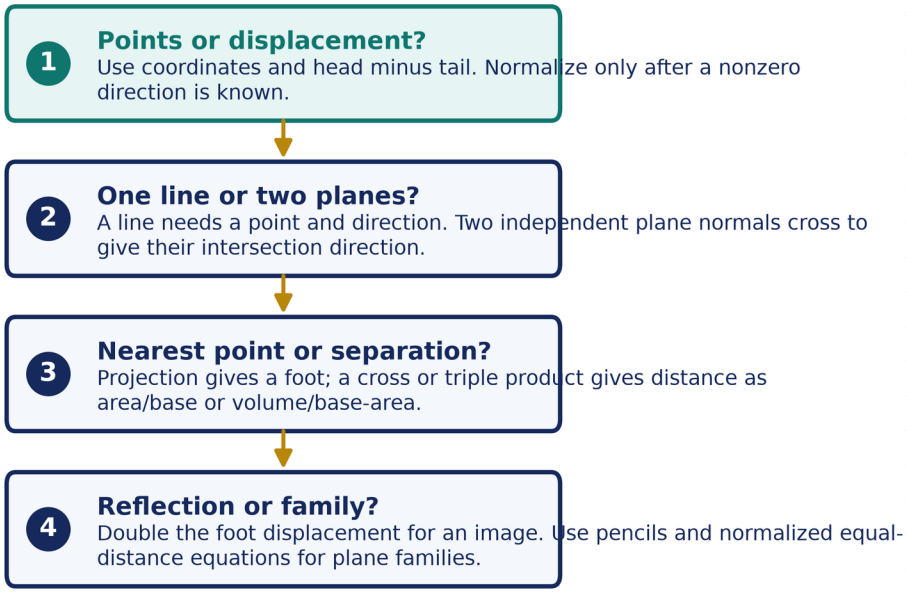
KEY RESULT

Second read: a parallel-plane distance uses the same normal

The plane $x+y+z=3$ contains P and is parallel to the original plane. Their separation is $|3-0|/\sqrt{3}=\sqrt{3}$. The foot F is therefore the nearest point of the original plane to the entire parallel plane through P.

Let the geometry choose the formula

A visual decision sequence for 3D problems



READ THE FIGURE

The picture identifies the object and the shortest reliable computation. Exact substitution then certifies the result.

Match each 3D question to its controlling quantity

Question	Computation	Essential check
Line direction	B-A or n1×n2	nonzero; perpendicular to normals
Two lines coplanar?	(a2-a1)·(b1×b2)	zero test after parallel test
Line and plane angle	|b·n|/(|b||n|)=sin(theta)	complement of line-normal angle
Point and plane distance	|p·n-d|/|n|	absolute normalized residual
Skew-line distance	|triple|/|b1×b2|	common perpendicular direction

READ THE FIGURE

Every formula has a hypothesis. Parallel directions make b1×b2 zero, so that case needs the point-to-line distance method instead.

Extension I — Sphere Foundations

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

LESSON 01 / 05

A sphere is a fixed-distance locus

With centre $C=(h,k,l)$ and radius $R>0$,

$$(x-h)^2+(y-k)^2+(z-l)^2=R^2.$$

Expanding and renaming coefficients gives

$$x^2+y^2+z^2+2ux+2vy+2wz+d=0,$$

with

$$C=(-u,\,-v,\,-w),\qquad R^2=u^2+v^2+w^2-d.$$

The sign of R^2 classifies the locus: positive gives a real sphere, zero a single point, and negative no real points.

WORKED EXAMPLE

Complete squares instead of guessing signs

For

$$x^2+y^2+z^2+2x-4y+6z-11=0,$$

move the constant and complete three squares:

$$(x+1)^2+(y-2)^2+(z-3)^2=25.$$

Thus the centre is $(-1,2,-3)$ and the radius is 5. Equivalently, $(u,v,w)=(1,-2,3)$ and $R^2=1+4+9-(-11)=25$.

The diameter form is a perpendicularity condition

If AB is a diameter, every other point P of the sphere satisfies $\angle APB=90^\circ$. Hence

$$(P-A)\cdot(P-B)=0.$$

The centre is the midpoint of AB and the radius is $AB/2$.

WORKED EXAMPLE

A sphere from two diameter endpoints

For $A=(1,0,0)$ and $B=(0,2,0)$,

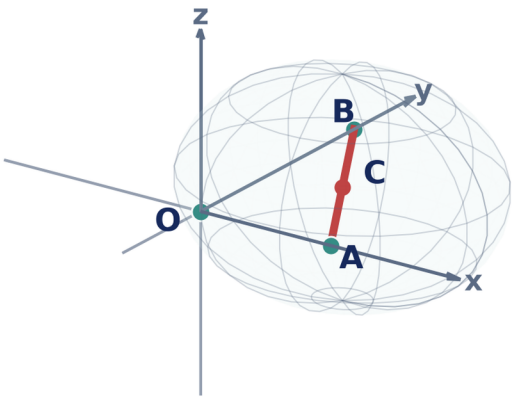
$$C=\left(\tfrac{1}{2},1,0\right), \qquad R=\tfrac{\sqrt{5}}{2}.$$

The equation is

$$\left(x-\tfrac{1}{2}\right)^2+(y-1)^2+z^2=\tfrac{5}{4},$$

or $x^2+y^2+z^2-x-2y=0$. The origin, A and B all satisfy it.

The midpoint of a diameter is the sphere's centre



READ THE FIGURE

C=(1/2,1,0), AB=sqrt(5), and R=sqrt(5)/2. The origin also lies on the sphere because OA is perpendicular to OB.

Four noncoplanar points determine a unique sphere

Substitute the four points into the general equation. Subtracting one equation from the other three eliminates d and leaves three linear equations for the centre coordinates. Noncoplanarity guarantees uniqueness.

WORKED EXAMPLE

A four-point sphere with exact centre and radius

For

$$A=(1,0,0),\quad B=(0,2,0),\quad C=(0,0,3),\quad D=(1,1,1),$$

equal squared distances from $O=(x,y,z)$ give

$$-2x+4y=3,\qquad -2x+6z=8,\qquad y+z=1.$$

Solving,

$$O=\left(-\tfrac{13}{10},\tfrac{1}{10},\tfrac{9}{10}\right).$$

Then

$$R^2=OA^2=\left(\tfrac{23}{10}\right)^2+\left(-\tfrac{1}{10}\right)^2+\left(-\tfrac{9}{10}\right)^2=\tfrac{611}{100}.$$

Thus $R=\sqrt{611}/10\approx 2.4718$. The other three distance checks agree.

COMMON TRAP

Centre signs and the constant term are independent

In the general equation, the centre is $(-u,-v,-w)$, while $d=u^2+v^2+w^2-R^2$. Do not read the centre directly from the linear coefficients without the minus signs, and do not use R^2 itself as the constant d.

KEY RESULT

Second read: four equal squared distances

For the four-point example,

$$|OA|^2=|OB|^2=|OC|^2=|OD|^2=\tfrac{611}{100}.$$

After clearing denominators, the four squared-distance numerators are $529+1+81$, $169+361+81$, $169+1+441$ and $529+81+1$, all equal to 611. Thus the centre and radius are checked without solving the system again.

LESSON 02 / 05

Compare centre–plane distance with the radius

For a sphere of radius R and a plane at distance d from its centre:

- $d < R$: the section is a circle;
- $d = R$: the plane is tangent at one point;
- $d > R$: there is no real intersection.

For a secant plane, the circle's centre Q is the foot of the perpendicular from the sphere's centre C. Pythagoras gives its radius:

$$r^2 = R^2 - d^2.$$

The vector CQ is normal to the plane, while every radius of the section circle lies in the plane.

WORKED EXAMPLE

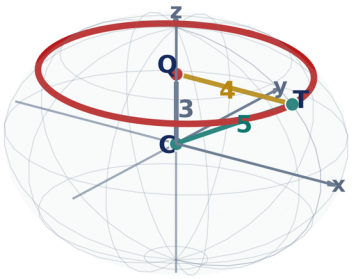
An axis-aligned section makes Pythagoras visible

The plane $z = 3$ meets $x^2 + y^2 + z^2 = 25$ in

$$x^2 + y^2 = 16, \qquad z = 3.$$

Its centre is $(0, 0, 3)$ and radius is 4. The sphere radius 5 is the hypotenuse of a 3-4-5 right triangle.

A plane section is a circle whose centre is the perpendicular foot



READ THE FIGURE

CQ=3 is perpendicular to the section plane, QT=4 lies in it, and CT=5 is a sphere radius. Hence 3^2+4^2=5^2.

WORKED EXAMPLE

A non-axis-aligned section with exact centre

For $2x - 3y + 4z = 20$ and $x^2 + y^2 + z^2 = 25$,

$$d = \frac{20}{\sqrt{29}} < 5.$$

The section centre is the foot from the origin:

$$Q = \tfrac{20}{29}(2, -3, 4) = (\tfrac{40}{29}, -\tfrac{60}{29}, \tfrac{80}{29}).$$

Its radius is

$$r = \sqrt{25 - \tfrac{400}{29}} = \sqrt{\tfrac{325}{29}} \approx 3.3477.$$

The tangent plane is perpendicular to the radius

At a point T on a sphere with centre C, every tangent direction is perpendicular to CT. Therefore the tangent plane is

$$(\mathbf{r} - \mathbf{t}) \cdot (\mathbf{t} - \mathbf{c}) = 0.$$

For $x^2 + y^2 + z^2 = R^2$, this reduces to $\mathbf{t} \cdot \mathbf{r} = R^2$.

WORKED EXAMPLE

Tangency checked by distance

At $T = (3, 0, 4)$ on $x^2 + y^2 + z^2 = 25$, the tangent plane is

$$3x + 4z = 25.$$

The origin is at distance $25/\sqrt{3^2 + 4^2} = 5$ from this plane, exactly the sphere radius. In contrast, $x + 2y + 2z = 9$ has distance $3 < 5$, so it cuts the sphere in a circle rather than touching it.

KEY RESULT

Section centre, section radius and tangent plane are linked

The foot formula supplies the section centre, $r^2 = R^2 - d^2$ supplies its radius, and $d = R$ is the boundary where the circle shrinks to the tangency point. Always compare d with R before declaring an intersection.

LESSON 03 / 05

Subtracting sphere equations leaves a radical plane

The power of P with respect to a sphere with centre C and radius R is $|P-C|^2-R^2$. For two spheres, subtracting their equations cancels all quadratic terms and gives the **radical plane**: the locus of points having equal power with respect to both. It is perpendicular to the line of centres. If the spheres intersect, their intersection circle lies in this plane.

WORKED EXAMPLE

A radical plane with a power check

For

$$x^2+y^2+z^2=25, \qquad (x-3)^2+y^2+z^2=16,$$

write the second as $x^2+y^2+z^2-6x-7=0$. Subtracting gives

$$6x-18=0 \quad\Longrightarrow\quad x=3.$$

At $(3,1,2)$, both powers are -11. The intersection circle is $y^2+z^2=16$ in the plane $x=3$.

Orthogonal spheres have perpendicular radii at intersection

Two intersecting spheres are orthogonal when their radii to a common point are perpendicular. If their centres are distance d apart, the triangle formed by the centres and that point gives

$$d^2=r_1^2+r_2^2.$$

In general-form coefficients this is equivalently

$$2(u_1u_2+v_1v_2+w_1w_2)=d_1+d_2.$$

WORKED EXAMPLE

Orthogonality checked two ways

For $x^2+y^2+z^2-6x=0$ and $x^2+y^2+z^2-6y=0$, the centres are $(3,0,0)$ and $(0,3,0)$, and both radii are 3. Hence

$$d^2=18=9+9=r_1^2+r_2^2.$$

The coefficient form gives $0=0$ as well. In contrast, radii $5,4$ with centre distance 5 do not satisfy orthogonality because $25\neq 25+16$.

Tangents from an external point form a cone

For P outside a sphere, let $D=PC>R$. Every tangent segment has length $\sqrt{D^2-R^2}$. The semi-vertical angle α of the tangent cone satisfies

$$\sin\alpha = \frac{R}{D}.$$

The contact points form a circle in the polar plane, at distance R^2/D from C along CP, with circle radius

$$\rho = R\sqrt{1-\frac{R^2}{D^2}}.$$

WORKED EXAMPLE

The 5–12–13 tangent cone

For $P=(13,0,0)$ and $x^2+y^2+z^2=25$,

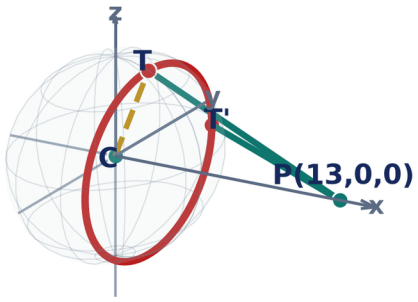
$$D=13, \qquad \sin\alpha=\frac{5}{13}, \qquad \alpha\approx 22.62^\circ.$$

The tangent length is 12. The contact circle lies in

$$x=\frac{25}{13}, \qquad y^2+z^2=\left(\frac{60}{13}\right)^2,$$

so its radius is $60/13$. Its plane is the polar plane $13x=25$ of P.

The contact circle lies in the external point's polar plane



READ THE FIGURE

CT=5 is perpendicular to PT=12, while CP=13. Every contact point lies on x=25/13 and has y^2+z^2=(60/13)^2.

Polar and radical constructions extend naturally

For $x^2+y^2+z^2=R^2$, the polar plane of $P=(x_1,y_1,z_1)$ is

$$xx_1+yy_1+zz_1=R^2.$$

For $P=(1,2,3)$ and $R=5$, it is $x+2y+3z=25$. Since P is inside the sphere, no real tangent cone exists, but the polar plane remains defined. If the polar of P contains Q, the polar of Q contains P.

WORKED EXAMPLE

Reciprocity checked on a contact point

For $P=(13,0,0)$ and $R=5$, take the contact point $T=(25/13,0,60/13)$. The polar of P is $13x=25$, which contains T. Conversely, the polar of T is

$$\tfrac{25}{13}x+\tfrac{60}{13}z=25,$$

and substituting P gives $25=25$. This verifies the reciprocal polar relation using the tangent construction.

Three spheres generally have three radical planes sharing a **radical axis**. For the source spheres $x^2+y^2+z^2=14$, $(x-5)^2+y^2+z^2=29$ and $x^2+(y-5)^2+z^2=19$, those planes give $x=1$, $y=2$ and $y=x+1$. Thus the radical axis is $(1,2,t)$, and every point on it has common power t^2-9.

KEY RESULT

A useful coaxal extension

A sphere orthogonal to two given spheres is orthogonal to every member of their coaxal pencil, and its centre lies on their radical plane. This follows by linearity of the general-form orthogonality condition.

LESSON 04 / 05

Volume and surface area determine the inradius

Join the incentre of a tetrahedron to its four vertices. This decomposes the solid into four pyramids whose heights are the inradius r. If V is the volume and S the total surface area,

$$V = \tfrac{1}{3} r S, \qquad r = \tfrac{3V}{S}.$$

With one vertex at the origin and the others at $\mathbf{a}, \mathbf{b}, \mathbf{c}$,

$$V = \tfrac{1}{6} |[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]|,$$

and four cross products give the face areas.

WORKED EXAMPLE

An exact inradius and incentre

For the tetrahedron

$$O = (0,0,0), \quad A = (4,0,0), \quad B = (0,4,0), \quad C = (0,0,3),$$

the coordinate-face areas are $8, 6, 6$, and the slant face has area $2\sqrt{34}$. Thus

$$S = 20 + 2\sqrt{34}, \qquad V = \tfrac{4 \cdot 4 \cdot 3}{6} = 8,$$

so

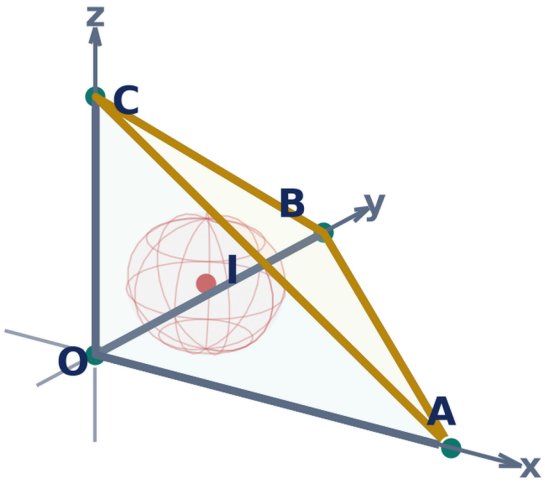
$$r = \frac{24}{20 + 2\sqrt{34}} = \frac{12}{10 + \sqrt{34}} \approx 0.7580.$$

The incentre is $I = (r, r, r)$: its distances to the three coordinate planes are r. The slant plane is $3x + 3y + 4z = 12$, and

$$\frac{12 - 10r}{\sqrt{34}} = r$$

is exactly the defining equation $12 = (10 + \sqrt{34})r$.

The insphere is tangent to all four faces



READ THE FIGURE
The rounded plotted radius is r=0.758; the exact value is 12/(10+sqrt(34)). Its centre I=(r,r,r) is equidistant from all four faces.

Monge's six midplanes concur

For each edge AB of a tetrahedron, form the plane through the midpoint of AB and perpendicular to the opposite edge CD. These six **Monge planes** are concurrent.

Place the circumcentre at the origin, so $|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = |\mathbf{d}| = R$, and define

$$\mathbf{m} = \frac{\mathbf{a}+\mathbf{b}+\mathbf{c}+\mathbf{d}}{2}.$$

For the midplane associated with AB,

$$\left(\mathbf{m} - \frac{\mathbf{a}+\mathbf{b}}{2}\right)\cdot(\mathbf{d}-\mathbf{c}) = \frac{\mathbf{c}+\mathbf{d}}{2}\cdot(\mathbf{d}-\mathbf{c}) = \frac{|\mathbf{d}|^2 - |\mathbf{c}|^2}{2} = 0.$$

Cyclic relabelling proves all six concurrences. In general coordinates,

$$\mathbf{m} = 2\mathbf{g} - \mathbf{o},$$

where $\mathbf{g}$ is the centroid and $\mathbf{o}$ the circumcentre. Thus the Monge point is the reflection of the circumcentre in the centroid.

WORKED EXAMPLE

An exact Monge point for a nonregular tetrahedron

For

$$A=(1,0,0),\quad B=(0,2,0),\quad C=(0,0,3),\quad D=(2,1,4),$$

solving equal distances to the four vertices gives

$$O=\left(\tfrac{47}{34},\tfrac{49}{34},\tfrac{61}{34}\right).$$

The centroid is $G=(3/4,3/4,7/4)$, so

$$M=2G-O=\left(\tfrac{2}{17},\tfrac{1}{17},\tfrac{29}{17}\right).$$

Substitution into all six midplane equations confirms the concurrence exactly.

SHORT DERIVATION

One of the six midplane checks

For edge AB, the midpoint is $(1/2,1,0)$ and the opposite direction is $D-C=(2,1,1)$. Hence

$$M-\left(\tfrac{1}{2},1,0\right)=\left(-\tfrac{13}{34},-\tfrac{16}{17},\tfrac{29}{17}\right).$$

Its dot product with $(2,1,1)$ is $(-26-32+58)/34=0$. The other five checks use the same perpendicularity condition.

LESSON 05 / 05

Spherical excess measures area

A spherical triangle is bounded by arcs of three great circles. Its angles are the angles between the corresponding great-circle planes. On a sphere of radius R, Girard's theorem gives

$$E = A + B + C - \pi > 0, \qquad \text{area} = R^2 E.$$

Angles must be in radians in this formula. The fraction of the whole sphere covered is $E/(4\pi)$.

WORKED EXAMPLE

Area from three spherical angles

On a unit sphere, angles $80^\circ, 70^\circ, 60^\circ$ give

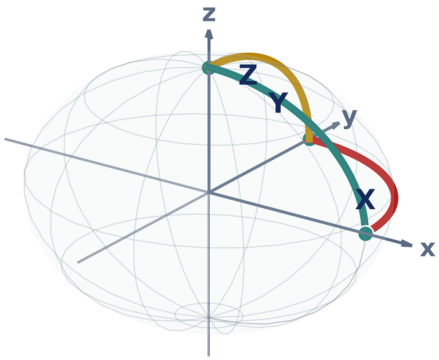
$$E = (80 + 70 + 60 - 180)^\circ = 30^\circ = \frac{\pi}{6}.$$

Thus the area is $\pi/6 \approx 0.5236$, and the fraction of the sphere is

$$\frac{\pi/6}{4\pi} = \frac{1}{24}.$$

If a sphere of radius 2 has spherical-triangle area 1, its excess is $1/4$ radian.

One octant is a spherical triangle with three right angles



READ THE FIGURE

The three arcs lie in perpendicular coordinate planes. E=3(pi/2)-pi=pi/2, so the area is pi/2, exactly one eighth of 4 pi.

A regular tetrahedron from symmetric coordinates

Take

$$A = (1, 1, 1), \quad B = (1, -1, -1), \quad C = (-1, 1, -1), \quad D = (-1, -1, 1).$$

Every edge has length $2\sqrt{2}$. The centroid and circumcentre are the origin, so $R=\sqrt{3}$. The face BCD lies in $x+y+z=-1$, whose distance from the origin is $1/\sqrt{3}$; hence $r=1/\sqrt{3}$ and $R=3r$.

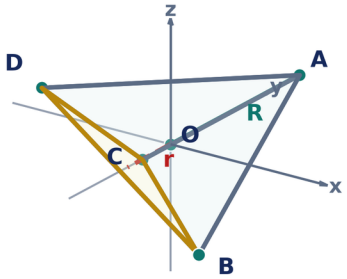
Scaling to edge length a gives

$$R=\frac{a\sqrt{6}}{4},\qquad r=\frac{a\sqrt{6}}{12},\qquad V=\frac{a^3}{6\sqrt{2}}.$$

The internal dihedral angle between any two faces is

$$\theta=\cos^{-1}\left(\tfrac{1}{3}\right)\approx 70.53^\circ.$$

In a regular tetrahedron, the circumradius is three times the inradius



READ THE FIGURE
For edge 2 sqrt(2), OA=R=sqrt(3). The foot to BCD is its centroid (-1/3,-1/3,-1/3), at distance r=1/sqrt(3).

WORKED EXAMPLE

Recover every measurement from the inradius

If $r=1$, then

$$a=\frac{12}{\sqrt{6}}=2\sqrt{6},\qquad R=3,$$

and

$$V=\frac{(2\sqrt{6})^3}{6\sqrt{2}}=8\sqrt{3}\approx 13.8564.$$

The solid angle uses dihedral angles, not face angles

At a vertex, a small sphere cuts out a spherical triangle whose **angles** are the tetrahedron's dihedral angles θ, while its **sides** are the face angles $\pi/3$. Girard's theorem therefore gives the solid angle

$$\Omega=3\theta-\pi=3\cos^{-1}\left(\tfrac{1}{3}\right)-\pi\approx 0.5513\ \text{sr}.$$